

Activity 15: Variable length subnet masks practice:

Challenge

In this activity, you will examine a network diagram. Having the ability to calculate the necessary number of subnets when presented with a network architecture is crucial.

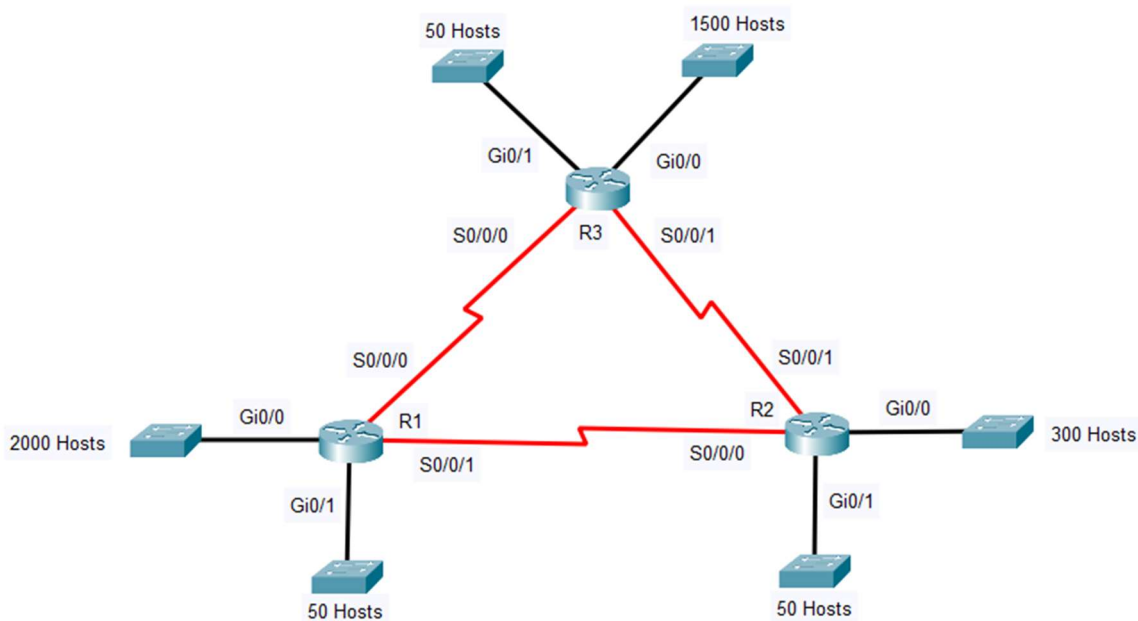
Not all organisations require subnets of equal size, thus subnetting may require more advanced techniques to ensure that network can operate efficiently using optimal IPv4 addressing.

You are required to calculate the number of bits to borrow, the number of hosts per subnet, and the possibility for expansion.

The organisation has a network address of 172.18.0.0/16 to be divided as illustrated in the following topology. You must choose an addressing scheme that can accommodate the number of networks and hosts in the topology.

You should begin by identifying the largest subnets and calculating the VLSM scheme to the small subnet requirements. This creates a simple hierarchical approach.

Topology



Instructions

Step 1: Determine the number of subnets in the network topology

1. How many subnets are there?

The topology requires subnets for:

2000 hosts

1500 hosts

300 hosts

50 hosts

Total subnets = 4

2. How many bits should you borrow to create the largest subnet of 2000 hosts?

Find the smallest number of host bits:

$$2^{11} - 2 = 2046$$

11 host bits are needed.

Assuming the original network is a Class B (/16), then:

$$16 - 11 = 5$$

Borrow 5 bits

3. How many bits should you borrow to create the subnet of 1500 hosts?

$$2^{11} - 2 = 2046$$

11 host bits are needed.

$$16 - 11 = 5$$

Borrow 5 bits

4. How many bits should you borrow to create the subnet of 300 hosts?

$$2^9 - 2 = 510$$

9 host bits are needed.

$$16 - 9 = 7$$

Borrow 7 bits

5. How many bits should you borrow to create the subnet of 50 hosts?

$$2^6 - 2 = 62$$

6 host bits are needed.

$$16 - 6 = 10$$

Borrow 10 bits

6. How many usable hosts (maximum) are supported on the subnet that must contain 2000 hosts?

$$2^{11} - 2 = 2046$$

2046 usable hosts

7. How many usable hosts (maximum) are supported on the subnet that must contain 1500 hosts?

$$2^{11} - 2 = 2046$$

2046 usable hosts

8. How many usable hosts (maximum) are supported on the subnet that must contain 300 hosts?

$$2^9 - 2 = 510$$

510 usable hosts

9. How many usable hosts (maximum) are supported on the subnet that must contain 50 hosts?

$$2^6 - 2 = 62$$

62 usable hosts

Step 2: Record the subnet information

Fill in the following table with the subnet information:

Subnet 0 should contain 2000 hosts.

Subnet 1 should contain 1500 hosts.

Subnet 2 should contain 300 hosts.

Subnet 3 should contain 50 hosts on R1.

Subnet 4 should contain 50 hosts on R2.

Subnet 5 should contain 50 hosts on R3.

Subnet 6 should contain R1 and R2.

Subnet 7 should contain R2 and R3.

Subnet 8 should contain R1 and R3.

Subnet number	Subnet address	First usable host address	Last usable host address	Broadcast address
0 (2000 hosts)	172.18.0.0/21	172.18.0.1	172.18.7.254	172.18.7.254
1 (1500 hosts)	172.18.8.0/21	172.18.8.1	172.18.15.254	172.18.15.254
2 (300 hosts)	172.18.16.0/23	172.18.16.1	172.18.17.254	172.18.17.254
3 (50 hosts - R1 LAN)	172.18.18.0/26	172.18.18.1	172.18.18.62	172.18.18.62
4 (50 hosts - R2 LAN)	172.18.18.64/26	172.18.18.65	172.18.18.126	172.18.18.126
5 (50 hosts - R3 LAN)	172.18.18.128/26	172.18.18.129	172.18.18.190	172.18.18.190
6 (R1-R2 link)	172.18.18.192/30	172.18.18.193	172.18.18.194	172.18.18.194
7 (R2-R3 link)	172.18.18.196/30	172.18.18.197	172.18.18.198	172.18.18.198
8 (R1-R3 link)	172.18.18.200/30	172.18.18.201	172.18.18.202	172.18.18.202

Step 3: Assign addresses to network devices in the subnets

Fill in the following table with IP addresses and subnet masks for the router interfaces:

Device	Interface	IP address	Subnet mask
R1	Gi0/0	172.18.0.1	255.255.248.0
	Gi0/1	172.18.18.1	255.255.255.192
	S0/0/0	172.18.18.193	255.255.255.252
	S0/0/1	172.18.18.201	255.255.255.252
R2	Gi0/0	172.18.8.1	255.255.248.0
	Gi0/1	172.18.18.65	255.255.255.192
	S0/0/0	172.18.18.194	255.255.255.252
	S0/0/1	172.18.18.197	255.255.255.252
R3	Gi0/0	172.18.16.1	255.255.254.0
	Gi0/1	172.18.18.129	255.255.255.192
	S0/0/0	172.18.18.198	255.255.255.252
	S0/0/1	172.18.18.202	255.255.255.252

Complete the following statements by filling in the blanks with the correct information:

A network containing 2000 hosts requires [blank] hosts [11] is leaves [21]
network bits remaining, expressed as prefix-length [blank] or a / 21 mask of
[255.255.248.0] requires the IP value in the third octet to increment in values of [blank].
[8]

A network containing 1500 hosts requires [blank] hosts [11] This leaves [blank] network [21]s
remaining, expressed as prefix-length / [blank] or a subnet mask / 21
[blank]. [255.255.248.0] formula requires the IP value in the third octet to increment in values
of [blank]. [8]

A network containing 300 hosts requires [blank] hosts [9] This leaves [blank] network [23]s
remaining, expressed as prefix-length / [blank] or a subnet mask / 23 mask of
[blank]. [255.255.254.0] formula requires the IP value in the third octet to increment in
values of [blank]. [2]

A network containing 50 hosts requires [blank] hosts. This leaves [blank] networks remaining, expressed as prefix-length /[blank] or a subnet mask of [blank]. 255.255.255.192 formula requires the IP value in the fourth octet to increment in values of [blank]. 64

A WAN link containing 2 hosts can require [blank] hosts. This leaves [blank] networks remaining, expressed as prefix-length /[blank] or a subnet mask of /30 [blank]. 255.255.255.252 formula requires the IP value in the fourth octet to increment in values of [blank]. 4

Note: On a point-to-point WAN link, it is also possible to implement /31 as a broadcast address is not required. See the following link for more information:

<https://www.ietf.org/rfc/rfc3021.txt>